Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 September 2026	**Team ID**	SWTID-2026-1595
Project Name	Import Data using Transform Maps (Spreadsheet)	**Maximum Marks**	4 Marks

Technical Architecture:

Spreadsheet (.xlsx) → **Employee Import** (Import Set staging table) → **Sample Spreadsheet Import** (Transform Map) → **Employee Test (u_employee_test)** → **Reports** → **Employee Analytics Dashboards**

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	ServiceNow platform interface used to create tables, load spreadsheet data, configure Transform Maps, reports and dashboards.	ServiceNow Web UI
2	Application Logic-1	Loads structured employee spreadsheet data into the staging Import Set table.	ServiceNow Import Sets
3	Application Logic-2	Maps source fields from Employee Import to Employee Test and executes the transformation.	ServiceNow Transform Maps
4	Application Logic-3	Prevents duplicate employee records and updates existing records using Employee ID matching.	Coalesce
5	Database	Stores employee records in the custom target table with Employee ID, Employee Name, Email, Department and Location.	ServiceNow Employee Test (u_employee_test)
6	Cloud Database	The project uses the ServiceNow platform data store for the application tables; no separate cloud database is specified in the source project.	ServiceNow platform data store
7	File Storage	Excel spreadsheet is used as the external source file for employee data import.	Excel (.xlsx) / Local file
8	External API-1	No external API is specified in the project.	Not specified
9	External API-2	No external API is specified in the project.	Not specified
10	Machine Learning Model	No machine learning model is used in the project.	Not applicable
11	Infrastructure (Server / Cloud)	Application and data import workflow are implemented on the ServiceNow platform.	ServiceNow platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The source project does not specify any open-source framework.	Not specified
2	Security	The project uses ServiceNow access controls for dashboard/report access; the source also mentions restricting Users, Groups and Roles from the Share option.	ServiceNow Users / Groups / Roles
3	Scalable Architecture	Bulk employee data is loaded through Import Sets and processed through Transform Maps, supporting repeated spreadsheet imports.	ServiceNow Import Sets & Transform Maps
4	Availability	Reports and dashboards provide a centralized view of employee records and import-related information within the ServiceNow platform.	ServiceNow Reports & Dashboards
5	Performance	Bulk spreadsheet import reduces the need for manual employee record entry and supports transformation of multiple rows in one import.	Import Sets & Transform Maps

Source basis: Entries are derived from the supplied ServiceNow Import Sets / Transform Maps project document. Items not specified in that document are marked accordingly.